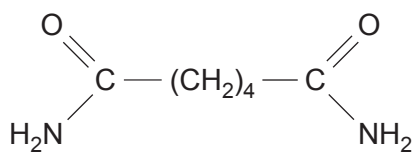
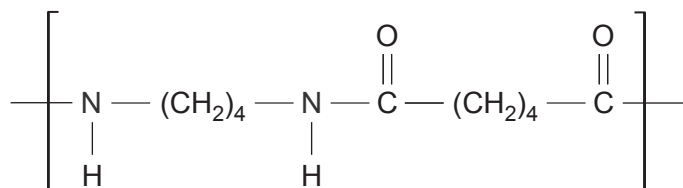


11. (a) A method to produce the polyamide **W**, starting from adipamide is shown below.



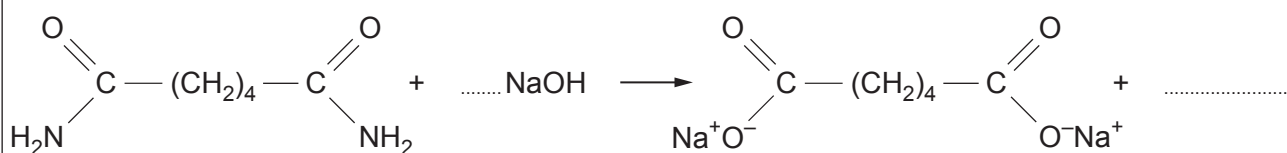
adipamide

polyamide **W**

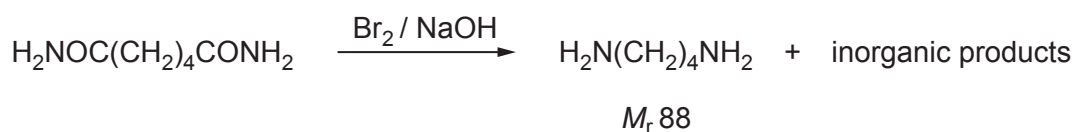
- (i) In the first stage adipamide is hydrolysed using aqueous sodium hydroxide, giving disodium hexanedioate as one of the products.

Complete and balance the equation for this stage.

[2]



- (ii) In another stage, adipamide is reacted with bromine in the presence of alkali to give butane-1,4-diamine, $\text{H}_2\text{N}(\text{CH}_2)_4\text{NH}_2$.



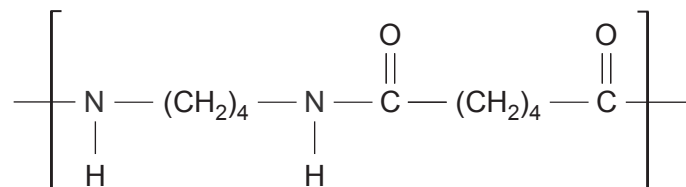
In this stage 0.075 mol of adipamide was used. If the percentage yield of the diamine was 60%, calculate the mass of butane-1,4-diamine produced.

[2]

Mass = g



- (iii) The sodium salt of hexanedioic acid produced in part (i) is then acidified and the hexanedioic acid produced is reacted with butane-1,4-diamine to give polyamide **W**.



Polyamides contain a peptide link.

Show the peptide link present in the polyamide **on the diagram above**. [1]

- (iv) An alternative method for producing hexanedioic acid is to treat adipamide with nitric(III) acid.



- I. Give **one** advantage of the method using nitric(III) acid. [1]

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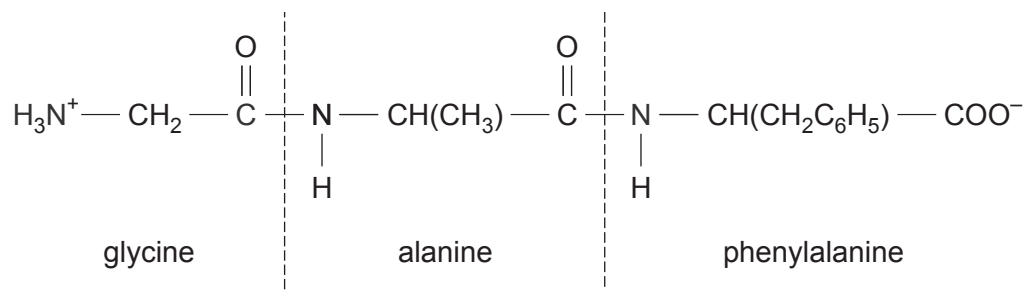
- II. Suggest how you would know that this alternative reaction using nitric(III) acid is complete. [1]

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- (b) A tripeptide is formed from three different amino acids.

Its formula (shown as a zwitterion) is



- (i) State why the formation of a peptide from amino acids is called a condensation reaction. [1]

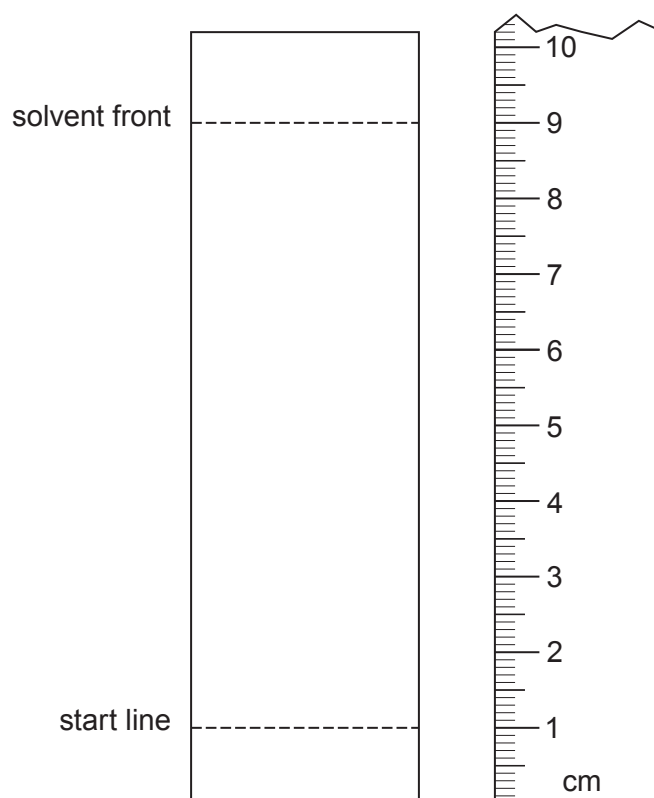
.....

- (ii) Write the formula of the species formed from phenylalanine in acid solution. [1]



- (iii) The tripeptide is hydrolysed to give its amino acid fragments. A solution containing these three amino acids is separated using thin layer chromatography.

Use the blank chromatogram to show the position of the 'spot' given by alanine, which has an R_f value of 0.54. Show how you arrived at your answer. [2]



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- (c) A solution contains a mass of **z g** of **one** of the two enantiomers of alanine. It rotates the plane of plane polarised light in the clockwise direction (+) by 6° ($+6^\circ$).

The other enantiomer rotates the plane of plane polarised light in the anticlockwise direction (–).

Another solution contains the same total mass **z g** of a **mixture** of both enantiomers. This mixture rotates the plane of plane polarised light by 1.6° in the anticlockwise direction (-1.6°).

Use this information to plot a graph to show how the rotation changes with the percentage of the (–) enantiomer. Hence determine the percentages of the (+) and (–) enantiomers in the mixture. [3]

Percentage of (+) enantiomer %

Percentage of (–) enantiomer %



Rotation / °

0

Percentage of (–) enantiomer / %

END OF PAPER

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